

5	(a) lost volts / energy used within the cell / internal resistance	B1	
	when cell supplies a current	B1	[2]

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- (b) (i) $E = I(R + r)$ C1
 $4.5 = 0.65 (6.0 + r)$
 $r = 0.92 \Omega$ A1 [2]
- (ii) $I = 0.65 \text{ (A)}$ and $V = IR$ C1
 $V = 0.65 \times 6 = 3.9 \text{ V}$ A1 [2]
- (iii) $P = V^2/R$ or $P = I^2R$ and $P = IV$ C1
 $= (3.9)^2/6 = 2.5 \text{ W}$ A1 [2]
- (iv) efficiency = power out/power in C1
 $= I^2R/I^2(R + r) = R/(R + r) = 6.0/(6.0 + 0.92) = 0.87$ A1 [2]
- (c) (circuit) resistance decreases B1
current increases M1
more heating effect A1 [3]